

# Maximal Tensor Products Preserve Operator-Algebra Quotients

A candidate full answer to a question in arXiv:0804.0028

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| <b>Status.</b> Candidate full solution, likely valid; human review pending. |
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## 1 Source question

Blecher and Duncan study nuclearity-related properties of nonselfadjoint operator algebras in [1]. Immediately before their Lemma 2.7 (Lemma 2.8 in the arXiv source file), they write:

“To deal with quotients we will need the following. We do not know if the result is true for nonselfadjoint  $D$ .”

Their lemma proves, when  $D$  is a  $C^*$ -algebra, that an approximately unital ideal  $A \triangleleft B$  satisfies

$$(B \otimes_{\max} D)/(A \otimes_{\max} D) \cong (B/A) \otimes_{\max} D$$

completely isometrically. It also proves the symmetric result when  $B$  is a  $C^*$ -algebra. The question is whether selfadjointness of either tensor factor is needed. The published erratum [2] does not alter this lemma or question.

To deal with quotients we will need the following. We do not know if the result is true for nonselfadjoint  $D$ :

**Lemma 2.7.** *If  $D$  is a  $C^*$ -algebra and if  $A$  is an approximately unital ideal in an operator algebra  $B$ , then  $(B \otimes_{\max} D)/(A \otimes_{\max} D) \cong (B/A) \otimes_{\max} D$  completely isometrically. The same relation holds if  $D$  is an approximately unital operator algebra and  $B$  is a  $C^*$ -algebra with ideal  $A$ .*

Figure 1: The question and Lemma 2.7 on page 4 of the source PDF.

## 2 Definitions

An operator algebra is a norm-closed, not necessarily selfadjoint algebra of operators on a Hilbert space. It is *approximately unital* if it has a contractive two-sided approximate identity. All ideals below are closed and two-sided.

The maximal operator-algebra tensor product  $E \otimes_{\max} F$  is characterized by commuting completely contractive representations: if  $\pi : E \rightarrow B(H)$  and  $\theta : F \rightarrow B(H)$  are commuting completely contractive homomorphisms, then

$$e \otimes f \longmapsto \pi(e)\theta(f)$$

extends completely contractively to  $E \otimes_{\max} F$ . Conversely, representations of the maximal tensor product have this factor form, with the usual passage to essential subspaces in the nonunital case; see [3, Chapter 6].

An isomorphism of operator algebras is *completely isometric* when it is isometric on every matrix level.

### 3 Proof intuition

The quotient map clearly induces a complete contraction from the left-hand quotient to  $(B/A) \otimes_{\max} D$ . The issue is the reverse matrix-norm inequality. Represent the left-hand quotient completely isometrically. Maximality splits the lifted representation into commuting factors  $\pi$  on  $B$  and  $\theta$  on  $D$ . Since tensors from  $A \otimes D$  vanish in the quotient,  $\pi(a)\theta(d) = 0$ . On the essential subspace generated by  $\theta(D)$  this forces  $\pi(a) = 0$ , so  $\pi$  factors through  $B/A$ . The two resulting factor representations then give a representation of  $(B/A) \otimes_{\max} D$  which recovers the original quotient representation without losing any matrix norm.

The essential-subspace step is the only point requiring care: it removes any hidden nondegeneracy assumption and is precisely what allows  $D$  to be nonselfadjoint.

### 4 The result

**Theorem 1.** *Let  $A$  be an approximately unital ideal in an operator algebra  $B$ , and let  $D$  be an approximately unital operator algebra. Then the canonical map*

$$\Phi : \frac{B \otimes_{\max} D}{A \otimes_{\max} D} \longrightarrow (B/A) \otimes_{\max} D, \quad \Phi((b \otimes d) + (A \otimes_{\max} D)) = (b + A) \otimes d,$$

*is a completely isometric isomorphism. In particular,*

$$0 \longrightarrow A \otimes_{\max} D \longrightarrow B \otimes_{\max} D \longrightarrow (B/A) \otimes_{\max} D \longrightarrow 0$$

*is an exact sequence of operator algebras.*

*Proof.* First,  $A \otimes_{\max} D$  embeds completely isometrically in  $B \otimes_{\max} D$ . For completeness, this is the ideal-extension argument of [1, Proposition 2.5]. If  $\pi : A \rightarrow B(H)$  and  $\theta : D \rightarrow B(H)$  are nondegenerate commuting completely contractive representations, the standard extension of  $\pi$  from the approximately unital ideal  $A$  to  $B$  is the completely contractive homomorphism  $\tilde{\pi}$  determined on the dense subspace  $\pi(A)H$  by

$$\tilde{\pi}(b)\pi(a)\xi = \pi(ba)\xi.$$

For  $a \in A$ ,  $b \in B$ ,  $d \in D$ , and  $\xi \in H$ ,

$$\theta(d)\tilde{\pi}(b)\pi(a)\xi = \theta(d)\pi(ba)\xi = \pi(ba)\theta(d)\xi = \tilde{\pi}(b)\theta(d)\pi(a)\xi.$$

Thus  $\tilde{\pi}$  commutes with  $\theta$ . The universal property of  $\otimes_{\max}$  gives the asserted complete isometry. Its image is a closed ideal.

Put

$$X = B \otimes_{\max} D, \quad I = A \otimes_{\max} D, \quad E = X/I,$$

and let  $p : X \rightarrow E$  and  $q : B \rightarrow B/A$  be the quotient maps. Functoriality of the maximal tensor product makes  $q \otimes \text{id}_D$  a completely contractive homomorphism from  $X$  to  $(B/A) \otimes_{\max} D$ . It annihilates  $I$  and therefore induces the completely contractive map  $\Phi$  in the theorem.

It remains to prove that  $\Phi$  is completely isometric. Choose a completely isometric representation  $\rho : E \rightarrow B(H)$ . By the factor-representation property of the maximal tensor product, there are commuting completely contractive homomorphisms

$$\pi : B \rightarrow B(H), \quad \theta : D \rightarrow B(H)$$

such that

$$\rho(p(b \otimes d)) = \pi(b)\theta(d) \quad (b \in B, d \in D). \quad (1)$$

Let

$$K = \overline{\theta(D)H}.$$

The space  $K$  is invariant for  $\pi(B)$  because  $\pi(B)$  commutes with  $\theta(D)$ . Moreover, for  $a \in A$  and  $d \in D$ , equation (1) gives

$$\pi(a)\theta(d) = \rho(p(a \otimes d)) = 0.$$

Since  $\theta(D)H$  is dense in  $K$ , it follows that  $\pi(a)|_K = 0$  for every  $a \in A$ . Consequently  $\pi|_K$  factors through a completely contractive homomorphism

$$\bar{\pi} : B/A \rightarrow B(K), \quad \bar{\pi}(b + A) = \pi(b)|_K.$$

The factor is completely contractive because  $q : B \rightarrow B/A$  is a complete quotient map. Also  $\theta_K(d) = \theta(d)|_K$  is a completely contractive representation of  $D$  on  $K$ , and it commutes with  $\bar{\pi}(B/A)$ . Maximality therefore gives a completely contractive representation

$$\sigma : (B/A) \otimes_{\max} D \rightarrow B(K), \quad \sigma((b + A) \otimes d) = \bar{\pi}(b + A)\theta_K(d).$$

For an algebraic tensor  $x \in B \otimes D$ , equations above show

$$\sigma(\Phi(p(x))) = \rho(p(x))|_K. \quad (2)$$

Every summand  $\pi(b)\theta(d)$  vanishes on  $K^\perp$ : if  $(e_t)$  is a contractive approximate identity for  $D$ , then  $\theta(e_t)$  converges strongly to the projection onto  $K$ , and  $\theta(d) = \theta(d)P_K$ . Its range is contained in  $K$ , since  $\pi(b)\theta(d) = \theta(d)\pi(b)$ . Hence restriction to  $K$  does not change the norm of  $\rho(p(x))$ , and the same assertion holds at every matrix level. By density, (2) holds for every  $u \in E$ . Therefore, for every  $n$  and  $u \in M_n(E)$ ,

$$\|u\| = \|\rho^{(n)}(u)\| = \|\sigma^{(n)}(\Phi^{(n)}(u))\| \leq \|\Phi^{(n)}(u)\|.$$

The reverse inequality follows from complete contractivity of  $\Phi$ . Thus  $\Phi$  is completely isometric.

Finally, the range of  $\Phi$  is closed and contains the algebraic tensor product  $(B/A) \otimes D$ , because  $q$  is onto. That algebraic tensor product is dense in  $(B/A) \otimes_{\max} D$ , so  $\Phi$  is onto.  $\square$

## 5 Verification notes

- The proof is matrix-level throughout: complete contractivity of  $\rho$ ,  $\bar{\pi}$ ,  $\theta_K$ , and  $\sigma$  gives the inequality in  $M_n$  for every  $n$ .
- Degenerate representations cause no loss. The tensor representation is zero on  $K^\perp$  and has range in  $K$ , so the compression used in the proof preserves norms exactly.
- No positivity or involution is used. The only properties of  $D$  are the maximal-tensor universal property and the existence of a contractive approximate identity.
- There is no computational component to verify; the claim is purely structural.

## 6 Scope, novelty check, and review recommendation

Theorem 1 gives the affirmative extension requested in the source: both  $B$  and  $D$  may be nonselfadjoint. Approximate unitality of  $D$  is the standing class used by the source and is needed for the clean essential-subspace argument.

A bounded search on 27 August 2026 covered the exact source title and authors, the phrase “nonselfadjoint  $D$ ”, close variants of “maximal tensor product quotient/projective”, the source’s citations, and the published erratum. It found Han’s projectivity theorem for the maximal *operator-system* tensor product [4], but no later paper explicitly answering this operator-algebra question and no exact occurrence of Theorem 1. Novelty is plausible rather than certified.

Human review should focus on the standard factor-representation property in the precise convention for  $\otimes_{\max}$  used in [1]. Once that defining property is confirmed, the remainder is an elementary quotient and essential-subspace argument.

## References

- [1] D. P. Blecher and B. L. Duncan, *Nuclearity-related properties for nonselfadjoint algebras*, J. Operator Theory **65** (2011), 47–70; arXiv:0804.0028.
- [2] D. P. Blecher and B. L. Duncan, *Errata to “Nuclearity-related properties for nonselfadjoint algebras”*, J. Operator Theory **65** (2011), 471–473.
- [3] D. P. Blecher and C. Le Merdy, *Operator Algebras and Their Modules: An Operator Space Approach*, London Mathematical Society Monographs 30, Oxford University Press, 2004.
- [4] K. H. Han, *On maximal tensor products and quotient maps of operator systems*, J. Math. Anal. Appl. **384** (2011), 375–386; arXiv:1010.0380.